

Ranjan Thakur

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About

Skilled in Python, JavaScript, and React, with experience building AI/ML systems and full-stack web applications. I enjoy solving real problems through clean code and continuous learning.

Education

Guru Gobind Singh Indraprastha University

September 2024 – present

B.tech in Artificial Intelligence and Machine Learning

- CGPA: 7.93/10.0
- **Coursework:** Data Structures & Algorithms, Digital Logic Design, Operating Systems, Database Management Systems, Probability & Statistics, Machine Learning, Web Technologies.

Projects

LiveTranslateSubs – Real-Time Speech Translation System

- Built a real-time Hindi/Hinglish-to-English subtitle system using faster-whisper (Medium model), FastAPI, and Socket.IO with GPU-accelerated inference on CUDA/float16
- Engineered an end-to-end browser-to-subtitle pipeline with AudioWorklet-based mic capture, VAD tuning, hallucination filtering, and SRT/VTT export support.

AI-BagCounter

- Developed a computer vision system for automated bag and luggage counting using a YOLO-based object detection model and OpenCV for real-time frame processing.
- Built a detection-to-count pipeline suitable for logistics and airport use cases, with bounding box visualization and per-frame count reporting.

RecruitDesk – AI Resume Ranking System

- Developed a full-stack AI-powered recruitment tool using FastAPI, React, spaCy, and sentence-transformers to semantically rank resumes against a given job description.
- Containerized the complete application using Docker, implementing an end-to-end pipeline from resume parsing and NLP extraction to ranked candidate output via a clean web interface.

AI-Powered Sign Language Translator

- Developed a real-time sign language recognition system using MediaPipe Holistic for keypoint extraction and an LSTM-based sequence model for gesture classification
- Implemented a live webcam inference pipeline in Python with OpenCV, achieving frame-by-frame gesture-to-text translation for accessibility applications.

Technologies

Languages Python, JavaScript, HTML, CSS

Libraries and frameworks: React.js, Node.js, Express.js, FastAPI, Socket.IO, NumPy, Pandas, scikit-learn, sentence-transformers, spaCy, OpenCV, MediaPipe, faster-whisper

Tools: Git, GitHub, VS Code, Docker, Google Colab, Notion

Achievements

- **President, ACE-DTC** (Nov 2025- Present): Leading technical events, workshops, and collaborative projects, fostering a developer community on campus.
- **Associate, ACE-DTC** (Jan 2025- Nov 2025): Content Writing , Graphic Designing, Writing Proposals and Handling Social Media